Spaced and digit formats. The spaced version of this cell shows the input as `C A B A | C C D A` and replaces "No spaces, no explanations." with "Write the output in the same format as the input: every character separated by a single space. No explanations."; everything else is unchanged. The digit version relabels A, B, C, D as $0, 1, 2, 3$ (input `2010|2230`), with the task line, rule, and mapping worded for digits ("Cyclic digit shift", "Replace every digit with the next one in the alphabet"). The same seed draws the same positions in all three formats.

`running_xor`, original and reset ($N = 8$, $k = 4$). Both prompts state the rule $y_i = x_i \oplus x_{i-1}$ with the first digit copied. The reset prompt adds: *"Immediately after each boundary, treat that position exactly like the first digit of the whole input—copy it as-is (nothing before a boundary counts as 'previous'), then resume the XOR rule from there until the next boundary,"* with the worked example `1011|0010` → `1110|0011`. For input `1101|1111`, the original expects `1011|0000`, while the reset variant expects `1011|1000`: only the digit after the marker differs.

`sliding_window_max` ($N = 8$, $k = 4$). The prompt states the rule with a worked example and adds: *"DO NOT include them [the bars] in the window or count them as digits."* For input `5170|1247` the expected output is `5577|7247`.

B Trials per cell

Table 9 gives trials per cell (one cell = task, model, length, density, and alphabet or modulus value) for the data pooled in Figures 2, 4, and 5. Gemini 3.5 Flash uses 3 on the letter and dependency tasks; Gemini 3.6 Flash uses 5 on all 13 task–format variants.

Table 9: Trials per cell, $N \leq 64$. Q2.5: Qwen2.5-Instruct; Q3.8: Qwen3.8-27B; Gemma 4: Gemma-4-31B-it.

Task	Q2.5-7B	Q2.5-14B	Q2.5-32B	Mistral	Phi	Q3.8	Gemma 4
`cyclic_shift`	20	20	20	40	20	20	20
`random_permutation`	40	40	40	40	20	20	20
`conditional_shift`	40	20	40	40	20	20	20
`copy_previous`	40	40	80	40	40	40	40
`running_xor`	40	40	40	40	20	40	40
`running_sum_local`	40	40	80	40	20	40	40
`sliding_window_max`	20	20	40	20	20	20	20
Reset variants	80	–	20	–	–	20	20
Spaced versions (each)	20	20	20	20	20	20	20
Digit versions (each)	20	20	20	20	20	20	20

Checkpoints. Hugging Face revisions: Qwen2.5-7B-Instruct `a09a35458c70`, Qwen2.5-14B-Instruct `cf98f3b3bbb4`, Qwen2.5-32B-Instruct `5ede1c97bbab`, Mistral-7B-Instruct-v0.3 `c170c708c41d`, Phi-3.5-mini-instruct `2fe192450127`, Qwen3.8-27B `37561816e66b`, Gemma-4-31B-it `842da3794eaa`.

Generation budget. The no-marker run and every spacing $k \geq 2$ share one budget, $1.5N + 20$ new tokens; $k = 1$ uses $2N + 20$. To check that the budget does not drive any effect, we re-tokenized every stored raw output of the seven open models ($N \leq 64$; 86,400 generations) and counted those that used the whole budget while giving fewer than N answer characters on the first line. Raw generations are stored for every spaced-letter and digit control run, every Qwen3.8 and Gemma 4 run, and, at Qwen2.5-32B, the second `sliding_window_max` batch and the $N = 128$ runs; the 80-trial second batches of `copy_previous` and `running_sum_local` at 32B (Table 9) do not store it. The original contiguous-letter and dependency grids of the five earlier models (Qwen2.5-7B, 14B, and the first 32B batch, Mistral, and Phi) predate raw-generation logging and retain only the clipped answer, which is why some marker-compliance entries elsewhere are unavailable. That count is an upper bound on cut-off answers, because it also counts short answers followed by unrequested text. The share is 0.58% without markers, 0.49% at $k \geq 2$, and 0.31% at $k = 1$; at the four

stronger models it is 0.04%, 0.02%, and 0%. Nearly all cases are Phi-3.5-mini on spaced letters (up to 19% of one cell without markers, against 14% with), and on the dependency tasks the share is 0.03% or less in every class. Because of the logging gap, the dependency-task compliance rows of Appendix L for Qwen2.5-7B, 14B, and 32B, Mistral, and Phi rest only on the runs that do store raw output: `sliding_window_max`, added later with logging on, and at Phi also `copy_previous`.

C How the uncertainty was computed

Pooled effects use the length-matched estimator of Section 3, with analytic standard errors over generations. Two checks ask how much that choice matters (`scripts/analyze_statistics_robustness.py`; per-cell values in `results_clean/_summary/statistics_robustness.csv`). Both resample 2,000 times, over the 91 pooled task–model cells of the three position-local families and the dependency family at the seven open models. The 125 cells of the main analysis also include the 15 reset cells and the 19 Gemini cells (6 at Gemini 3.5 with 3 trials each, 13 at Gemini 3.6 with 5), which are not resampled.

Resampling generations reproduces the analytic result. Bootstrapping generations within each condition and length gives a median $|z|$ within 3% of the analytic value, and the Bonferroni verdict ($|z| \geq 3.54$) changes in 1 of 91 cells. The analytic standard errors are therefore not the weak point.

Resampling whole cells is far wider. Treating each (length, spacing, alphabet) cell as the unit instead, so that whole cells enter or leave a resample, widens the median interval from 0.046 to 0.155 and leaves 16 of 91 cells above the Bonferroni threshold instead of 62. Two things drive this, and neither is a defect in the data. First, a pooled cell deliberately spans lengths at which the effect is absent: at $N \leq 16$ these models are near ceiling and markers do nothing, which is itself Finding 2. Averaging those in and then asking how much the average moves when lengths are resampled mostly measures that by-design heterogeneity. Second, a pooled cell contains only 10 to 25 (length, spacing, alphabet) cells, and a bootstrap over so few clusters has little power.

Restricting to the length where the claim is actually made removes the first problem. At $N = 64$, all 24 spaced-letter and digit cells at the four stronger models have a positive effect, between +0.07 and +0.37 (mean +0.24); all 24 clear $|z| \geq 3.54$ analytically and under the generation bootstrap, 21 of 24 have a cell-level $|z| \geq 2$, and 4 of 24 clear 3.54 under the cell-level bootstrap with its 15 or so clusters (`results_clean/_summary/finding2_at_N64.csv`). Both bootstraps use a fixed seed (20261111).

We therefore read a pooled z as a statement about the mean effect over the grid we ran, not as evidence that the effect is uniform across lengths and spacings, nor that it would survive a different choice of grid. The claims about structure rest on the length-resolved and position-resolved analyses (Figures 4 and 3), not on the pooled numbers.

The baseline's marker sentence. The no-marker prompt keeps the sentence that describes the markers. To measure what that costs, we reran the no-marker condition at the four stronger models with every mention of markers removed from the prompt, on the same inputs (`scripts/run_clean_baseline.py`): the ten task variants of the two tokenization controls and the dependency family, $N \in \{16, 32, 64\}$, every alphabet or modulus value, 20 trials. Table 10 gives the paired difference (clean minus with-sentence) per model and family. It is +0.006 overall (95% interval $[0.002, 0.009]$ over 300 cells), between −0.015 and +0.026 in every model–family combination, and negative, meaning the sentence helped, in 105 of the 300 cells. Recomputing Finding 2 at $N = 64$ against the clean baseline moves the share of remaining errors removed by at most 0.014 at any model.

Dropping the redundant configurations. At $A = 2$ the three position-local rules coincide and at $M = 2$ `running_sum_local` computes `running_xor` (Section 3). Recomputing all 91 pooled cells without those configurations moves the effect by 0.013 in the median cell and at most 0.054; the effect gets *larger* in 54 of 91 cells, because the two-symbol cells are the easiest and dilute the average. Three cells change sign, all with $|\Delta| \leq 0.009$ and none significant either way, and the Bonferroni verdict changes in 5 of 91 cells. Finding 2 is unchanged: 23 of its 24 cells remain significant.

Table 10: Cost of the baseline's marker sentence: no-marker accuracy with the sentence removed minus with it present, on the same inputs, mean over (task, length, alphabet) cells with a 95% interval over cells. $N \in \{16, 32, 64\}$, 20 trials per cell.

Model	Spaced letters	Digits	Dependency	All
Qwen2.5 14B	$+0.011 \pm 0.006$	$+0.005 \pm 0.007$	-0.015 ± 0.015	$+0.002 \pm 0.006$
Qwen2.5 32B	-0.002 ± 0.009	-0.002 ± 0.010	$+0.015 \pm 0.011$	$+0.003 \pm 0.006$
Qwen3.8 27B	-0.009 ± 0.022	$+0.017 \pm 0.013$	$+0.014 \pm 0.011$	$+0.007 \pm 0.010$
Gemma 4 31B	$+0.007 \pm 0.010$	$+0.005 \pm 0.010$	$+0.026 \pm 0.013$	$+0.012 \pm 0.007$
All				$+0.0057 \pm 0.0036$

Paired inputs. The main grids draw a fresh input for every (length, spacing, trial) cell, so the no-marker and marker runs of a trial transform different strings and every effect is an unpaired difference of means. To measure what that costs, we reran the spaced-letter and digit controls and the four dependency tasks at $N \in \{16, 32, 64\}$ with the input of each trial fixed across spacings (`scripts/run_paired_inputs.py`; 20 trials; Table 11). The paired and unpaired estimates of the effect agree to within 0.01 in every model and format, the correlation between a trial's marker and no-marker accuracies is 0.11 to 0.28, and the paired standard error of a cell is 83% to 94% of the unpaired one, so pairing would have tightened the intervals by 6% to 17%. The paired run's no-marker rows reuse the main grid's seeds and reproduce its outputs exactly in 84% to 100% of generations, depending on the model.

Table 11: Paired-input rerun. Per model and format, averaged over cells (task, alphabet or modulus, N, $k > 0$): the marker effect from the main grid (different inputs per condition) and from the paired run (same inputs), the mean standard error of a cell under the unpaired and the paired estimator, and the mean correlation r between a trial's marker and no-marker accuracies.

			Δ		Cell SE		
Model	Format	Cells	Main grid	Paired	Unpaired	Paired	r
Qwen2.5 7B	Spaced letters	126	+0.041	+0.037	0.050	0.044	0.21
	Digits	126	+0.047	+0.052	0.045	0.038	0.26
	Dependency	98	−0.010	−0.005	0.042	0.035	0.26
Mistral 7B	Spaced letters	126	+0.058	+0.056	0.038	0.034	0.16
	Digits	126	−0.015	−0.008	0.043	0.039	0.18
	Dependency	31	−0.006	−0.008	0.040	0.038	0.11
Phi-3.5 mini	Spaced letters	93	+0.014	+0.023	0.042	0.035	0.28

D Per-model results

Figures 7 and 8 show accuracy by sequence length for each open model and Gemini 3.6 Flash by task; Gemini 3.5 Flash is in Table 19. Pooled effects and their significance are in Figure 5.

Mistral-7B. In all three position-local tasks, the benefit is largest at short lengths and decays to zero at the longest length tested (`cyclic_shift`, $N = 256$: $+0.008$, $z = 0.4$; `random_permutation`, $N = 64$: -0.009, $z = 0.5$; `conditional_shift`, $N = 64$: -0.006, $z = 0.4$). In Qwen models the benefit holds or grows with length.

Repetition collapse. On `running_xor`, Qwen2.5-7B repeats a single wrong digit in about 38% of trials, more often under dense markers. This failure falls to 0.9% at 14B and 0.1% for Mistral-7B, so it is specific to Qwen2.5-7B.

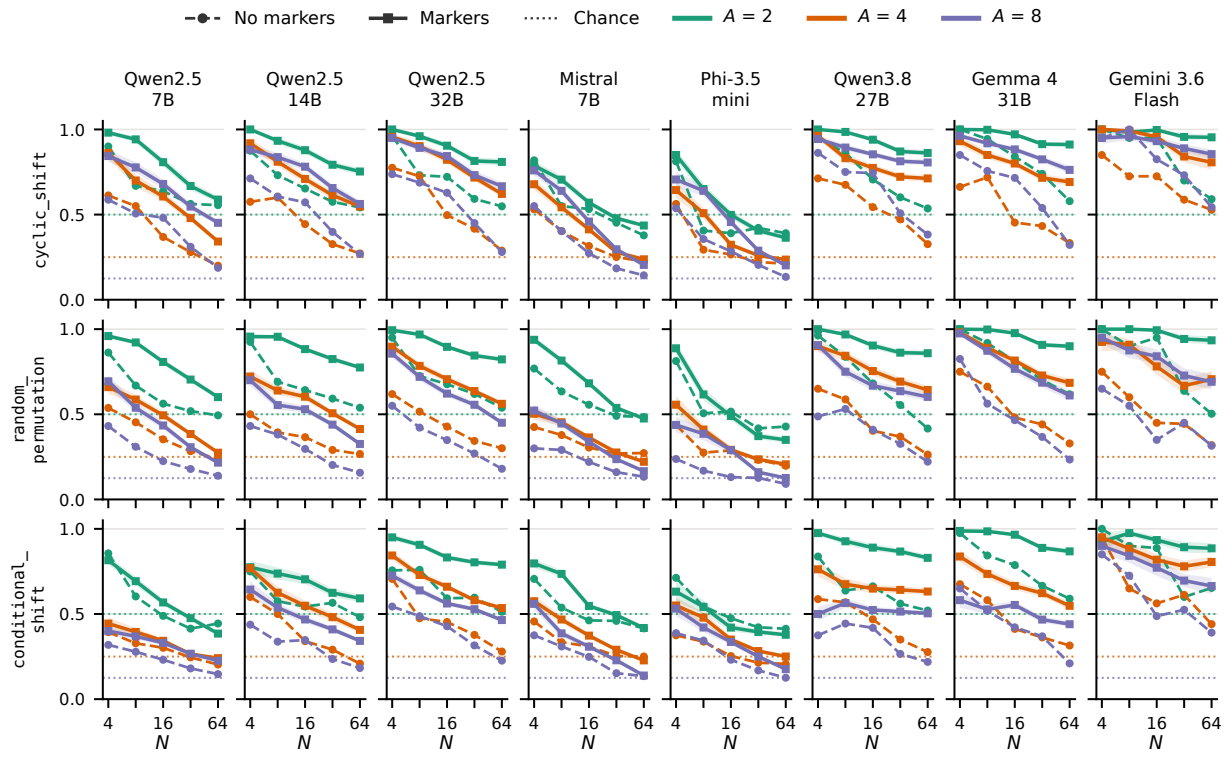


Figure 7: Position-local tasks on contiguous letters, by model and length. Character accuracy against sequence length N (4 to 64). Dashed: no markers; solid: markers (mean over spacings $k \in \{1, 2, 4, 8, 16\}$ with $k < N$); colors: alphabet size; dotted: chance; bands: ± 1 standard error.

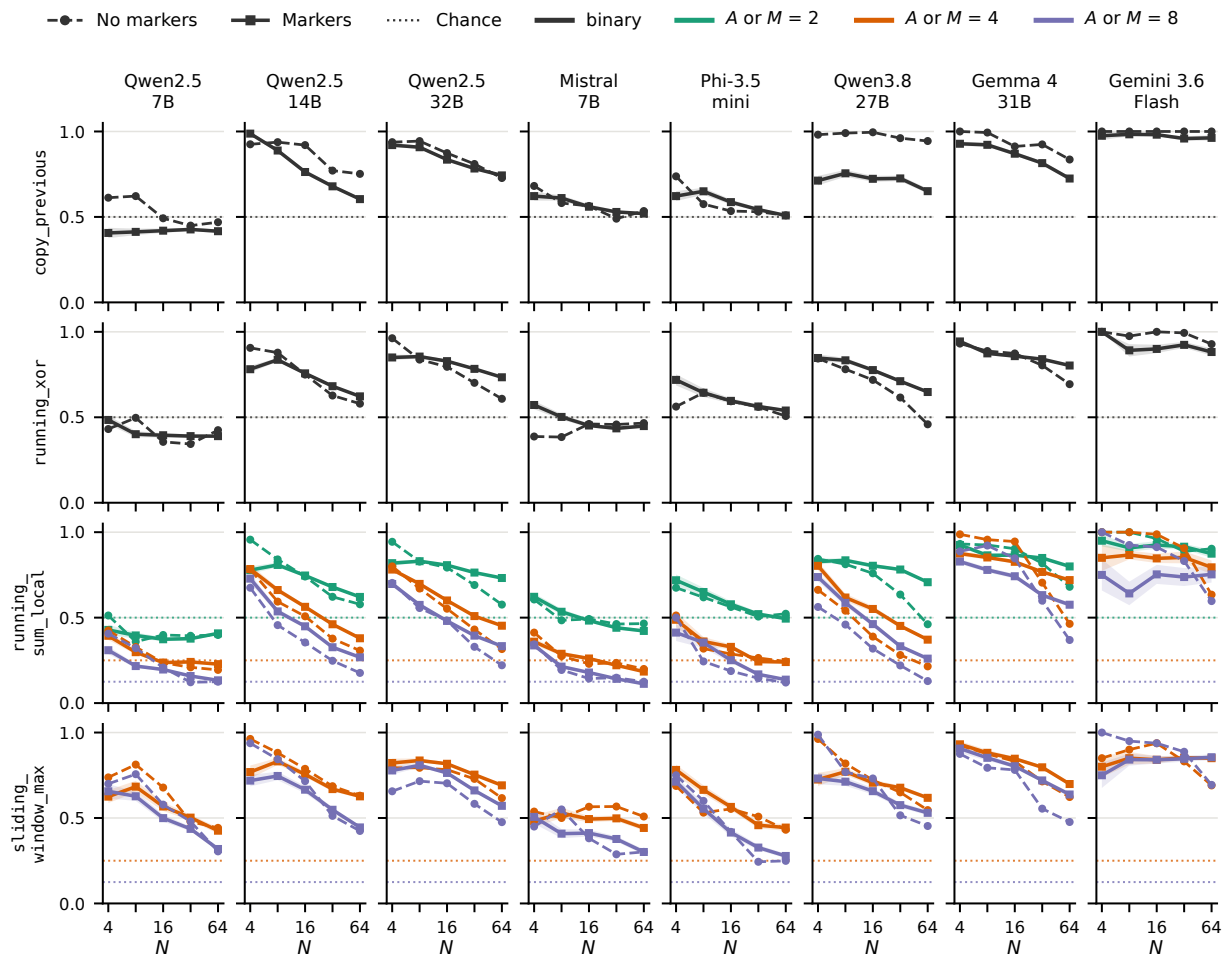


Figure 8: Dependency tasks (digits), by model and length; same conventions as Figure 7. Colors: modulus (`running_sum_local`) or alphabet size (`sliding_window_max`); black: binary. Marker runs average over all positions, so they mix the post-marker loss with the within-segment gain (Section 6).

Out-of-alphabet answers. On the digit tasks, Qwen2.5-7B often writes a letter (usually `t`, sometimes `o`) in place of `0`, in 34% to 39% of outputs, with or without markers. On spaced letters, several models step outside the alphabet (for example `I` after `H` instead of wrapping to `A`). Such characters are scored as errors.

E Per-position accuracy

Table 12 gives the per-position means plotted in Figure 3, and Table 13 breaks them down by distance from the marker. Table 14 repeats the dependency-task comparison under alignment-robust scoring, Table 16 repeats it at $k = 2$, 3, and 4 separately, and Figure 9 shows the length profile of the tokenization controls for each task. "No markers" is the no-marker run's accuracy at the same position indices. Means are over the tasks in each block; per-task values are in `results_clean/_summary/per_position_values.csv`.

Error clustering. Table 15 asks whether markers change how errors cluster along the sequence. For every generation at $N = 64$ of the four stronger open models, the per-position error indicator is computed under index scoring and under the edit-distance alignment, and three quantities are pooled over generations: the error rate, the probability $p_{1|1}$ that the position after an error is also wrong, and the probability $p_{1|0}$ that the position after a correct one is wrong. Under index scoring a dropped or inserted character makes every later position wrong, so $p_{1|1}$ there measures cascading misalignment; under alignment it measures the clustering of genuine local errors. Without markers the index-scored $p_{1|1}$ is 0.70 to 0.77 on the spaced and digit formats; sparse markers ($k \in \{2, 4, 8, 16\}$) bring it to 0.24 to 0.59, and the mean run of consecutive errors shortens from 3.1 to 3.9 positions to 1.3 to 2.3. Under alignment the change is smaller (0.34 to 0.51 against 0.21 to 0.40) and $p_{1|0}$ falls together with the error rate, so sparse markers mostly stop a slip from propagating, with a smaller reduction in the clustering of substitutions. A marker after every character lowers the error rate but does not shorten the cascades (index-scored $p_{1|1}$ of 0.45 to 0.70). On the dependency tasks the same shift appears (0.63 to 0.79 against 0.40 to 0.62, index-scored), on top of the errors the blind spot adds at the post-marker positions. Script: `scripts/analyze_error_autocorrelation.py`.

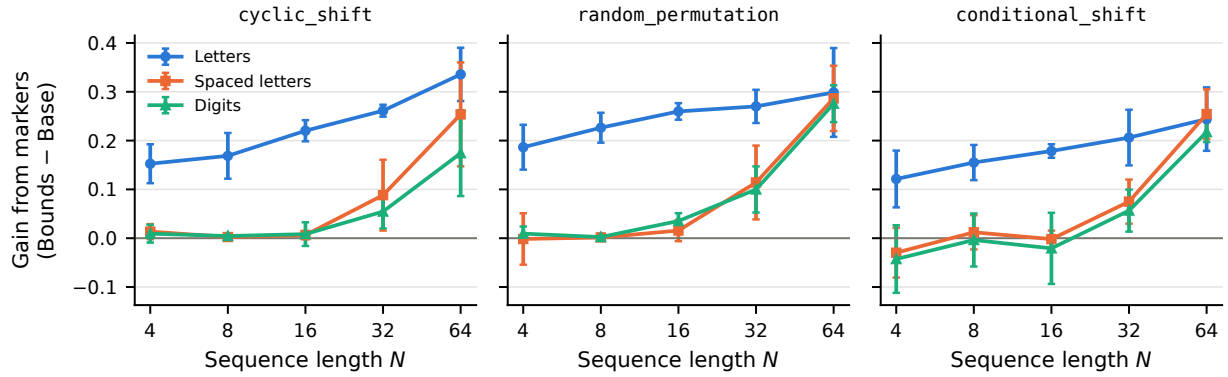


Figure 9: The length profile of Figure 4a, task by task (mean over Qwen2.5-14B and 32B, Qwen3.8-27B, and Gemma-4-31B; whiskers are 95% intervals over models). All three tasks show the same pattern.

Table 12: Per-position accuracy by model ($N \leq 64$, $k \in \{2, 4, 8, 16\}$), mean over the tasks in each block.

	Within a segment			Right after a marker		
Model	No markers	Markers	Δ	No markers	Markers	Δ
Position-local tasks, letters						
Qwen2.5 7B	0.38	0.45	+0.07	0.37	0.51	+0.14
Qwen2.5 14B	0.43	0.55	+0.11	0.40	0.63	+0.23
Qwen2.5 32B	0.49	0.65	+0.16	0.45	0.79	+0.34
Mistral 7B	0.35	0.35	0.00	0.30	0.39	+0.09
Phi-3.5 mini	0.31	0.31	0.00	0.28	0.35	+0.07
Qwen3.8 27B	0.50	0.69	+0.20	0.47	0.81	+0.34
Gemma 4 31B	0.56	0.72	+0.16	0.51	0.80	+0.29
Gemini 3.5 Flash	0.57	0.74	+0.17	0.52	0.83	+0.32
Gemini 3.6 Flash	0.63	0.80	+0.16	0.59	0.94	+0.34
Position-local tasks, spaced letters						
Qwen2.5 7B	0.54	0.55	+0.01	0.52	0.52	0.00
Qwen2.5 14B	0.69	0.80	+0.11	0.67	0.82	+0.15
Qwen2.5 32B	0.79	0.93	+0.14	0.79	0.92	+0.14
Mistral 7B	0.40	0.43	+0.03	0.35	0.45	+0.09
Phi-3.5 mini	0.35	0.35	−0.01	0.33	0.34	+0.01
Qwen3.8 27B	0.83	0.92	+0.09	0.84	0.92	+0.08
Gemma 4 31B	0.88	0.94	+0.06	0.89	0.94	+0.05
Gemini 3.6 Flash	0.96	0.99	+0.03	0.97	0.99	+0.03
Position-local tasks, digits						
Qwen2.5 7B	0.35	0.38	+0.04	0.30	0.35	+0.05
Qwen2.5 14B	0.77	0.85	+0.08	0.77	0.87	+0.09
Qwen2.5 32B	0.83	0.94	+0.11	0.83	0.93	+0.09
Mistral 7B	0.43	0.38	−0.04	0.41	0.39	−0.02
Phi-3.5 mini	0.48	0.55	+0.07	0.44	0.56	+0.12
Qwen3.8 27B	0.85	0.94	+0.09	0.85	0.94	+0.09
Gemma 4 31B	0.92	0.99	+0.07	0.92	0.98	+0.06
Gemini 3.6 Flash	0.99	1.00	+0.01	0.99	1.00	+0.01
Dependency tasks						
Qwen2.5 7B	0.43	0.41	−0.02	0.42	0.36	−0.06
Qwen2.5 14B	0.68	0.68	0.00	0.64	0.49	−0.15
Qwen2.5 32B	0.70	0.77	+0.07	0.66	0.52	−0.15
Mistral 7B	0.44	0.43	0.00	0.43	0.38	−0.04
Phi-3.5 mini	0.49	0.48	0.00	0.45	0.42	−0.03
Qwen3.8 27B	0.68	0.70	+0.02	0.63	0.44	−0.19
Gemma 4 31B	0.79	0.85	+0.05	0.78	0.52	−0.26
Gemini 3.5 Flash	0.92	0.97	+0.05	0.92	0.49	−0.43
Gemini 3.6 Flash	0.92	0.95	+0.02	0.91	0.57	−0.34
Dependency tasks, reset variants						
Qwen2.5 7B	0.43	0.43	0.00	0.41	0.58	+0.17
Qwen2.5 32B	0.69	0.79	+0.10	0.65	0.92	+0.27
Qwen3.8 27B	0.66	0.68	+0.03	0.63	0.98	+0.36
Gemma 4 31B	0.77	0.85	+0.08	0.75	0.97	+0.22

Table 13: Accuracy by distance after the marker at $k = 16$ ($N \in \{32, 64\}$): 0 is the position right after the marker. Mean over the tasks in each block.

		Positions after the marker				
Tasks	Model	0	1	2–3	4–7	8–15
Position-local, letters	Qwen2.5 7B	0.32	0.38	0.35	0.29	0.26
	Qwen2.5 14B	0.45	0.47	0.41	0.37	0.34
	Qwen2.5 32B	0.60	0.57	0.48	0.45	0.41
	Mistral 7B	0.28	0.34	0.29	0.26	0.25
	Phi-3.5 mini	0.30	0.28	0.28	0.28	0.23
	Qwen3.8 27B	0.68	0.67	0.59	0.51	0.47
	Gemma 4 31B	0.52	0.62	0.55	0.51	0.43
	Gemini 3.5 Flash	0.74	0.69	0.65	0.51	0.46
	Gemini 3.6 Flash	0.88	0.79	0.73	0.65	0.59
Position-local, spaced	Qwen2.5 7B	0.30	0.46	0.47	0.43	0.35
	Qwen2.5 14B	0.59	0.67	0.68	0.66	0.59
	Qwen2.5 32B	0.86	0.90	0.89	0.88	0.85
	Mistral 7B	0.31	0.39	0.37	0.27	0.24
	Phi-3.5 mini	0.25	0.37	0.33	0.25	0.21
	Qwen3.8 27B	0.89	0.88	0.91	0.91	0.90
	Gemma 4 31B	0.93	0.94	0.95	0.93	0.94
	Gemini 3.6 Flash	0.99	0.98	1.00	0.98	0.98
Position-local, digits	Qwen2.5 7B	0.28	0.38	0.32	0.30	0.25
	Qwen2.5 14B	0.83	0.85	0.86	0.81	0.76
	Qwen2.5 32B	0.90	0.93	0.94	0.93	0.90
	Mistral 7B	0.31	0.38	0.36	0.31	0.26
	Phi-3.5 mini	0.38	0.53	0.52	0.46	0.37
	Qwen3.8 27B	0.90	0.89	0.91	0.91	0.90
	Gemma 4 31B	0.94	0.98	0.99	0.98	0.98
	Gemini 3.6 Flash	0.99	0.99	0.99	0.99	0.99
Dependency	Qwen2.5 7B	0.36	0.41	0.42	0.42	0.36
	Qwen2.5 14B	0.45	0.75	0.68	0.60	0.53
	Qwen2.5 32B	0.49	0.85	0.79	0.71	0.63
	Mistral 7B	0.41	0.42	0.45	0.42	0.41
	Phi-3.5 mini	0.40	0.47	0.44	0.40	0.38
	Qwen3.8 27B	0.46	0.70	0.71	0.65	0.58
	Gemma 4 31B	0.49	0.82	0.83	0.81	0.75
	Gemini 3.5 Flash	0.57	1.00	1.00	1.00	0.97
	Gemini 3.6 Flash	0.49	0.91	0.97	0.95	0.93

Table 14: Per-position gains on dependency tasks (marker run minus the no-marker run at the same positions) under index-by-index scoring and under edit-distance alignment, which repairs dropped or inserted characters. $N \leq 64$, $k \in \{2, 4, 8, 16\}$.

		Right after a marker		Within a segment	
Task	Model	Index	Aligned	Index	Aligned
`copy_previous`	Qwen2.5 7B	−0.10	−0.09	−0.07	−0.06
	Qwen2.5 14B	−0.21	−0.15	−0.10	−0.05
	Qwen2.5 32B	−0.15	−0.17	−0.02	−0.02
	Mistral 7B	−0.05	−0.01	0.00	+0.01
	Phi-3.5 mini	−0.04	−0.07	−0.01	−0.04
	Qwen3.8 27B	−0.45	−0.20	−0.24	−0.05
	Gemma 4 31B	−0.29	−0.25	−0.07	−0.04
`running_xor`	Qwen2.5 7B	+0.02	−0.02	+0.01	0.00
	Qwen2.5 14B	−0.14	−0.13	+0.05	+0.01
	Qwen2.5 32B	−0.19	−0.24	+0.11	+0.03
	Mistral 7B	−0.04	−0.03	+0.02	−0.03
	Phi-3.5 mini	−0.05	−0.02	+0.03	+0.03
	Qwen3.8 27B	−0.06	−0.08	+0.13	+0.06
	Gemma 4 31B	−0.26	−0.29	+0.09	+0.04
`running_sum_local`	Qwen2.5 7B	−0.03	−0.04	−0.01	−0.02
	Qwen2.5 14B	−0.08	−0.12	+0.07	+0.03
	Qwen2.5 32B	−0.16	−0.22	+0.11	+0.05
	Mistral 7B	−0.01	−0.04	0.00	−0.01
	Phi-3.5 mini	−0.03	0.00	−0.02	−0.03
	Qwen3.8 27B	−0.06	−0.13	+0.14	+0.09
	Gemma 4 31B	−0.41	−0.46	+0.11	+0.07
`sliding_window_max`	Qwen2.5 7B	−0.11	−0.06	−0.02	−0.02
	Qwen2.5 14B	−0.16	−0.10	−0.03	−0.01
	Qwen2.5 32B	−0.09	−0.07	+0.08	+0.07
	Mistral 7B	−0.07	0.00	−0.02	0.00
	Phi-3.5 mini	+0.02	−0.03	−0.02	−0.03
	Qwen3.8 27B	−0.18	−0.16	+0.05	+0.04
	Gemma 4 31B	−0.07	−0.08	+0.08	+0.02

Table 15: Error clustering at $N = 64$ at the four stronger open models, pooled over the tasks of each format and over alphabets or moduli: error rate, $p_{1|1}$ (an error follows an error), and $p_{1|0}$ (an error follows a correct position), by marker density, under index scoring (top) and edit-distance alignment (bottom).

		No markers			$k = 1$			$k \in \{2, 4, 8, 16\}$		
Model	Format	rate	$p_{1\|1}$	$p_{1\|0}$	rate	$p_{1\|1}$	$p_{1\|0}$	rate	$p_{1\|1}$	$p_{1\|0}$
Index scoring										
Qwen2.5 14B	Spaced letters	0.59	0.76	0.36	0.35	0.70	0.17	0.28	0.59	0.17
	Digits	0.45	0.70	0.26	0.31	0.64	0.17	0.18	0.45	0.12
	Dependency tasks	0.51	0.70	0.34	0.51	0.66	0.36	0.49	0.61	0.39
Qwen2.5 32B	Spaced letters	0.47	0.74	0.25	0.22	0.69	0.10	0.11	0.55	0.06
	Digits	0.36	0.71	0.17	0.23	0.70	0.10	0.07	0.42	0.05
	Dependency tasks	0.51	0.69	0.34	0.42	0.63	0.28	0.40	0.55	0.32
Qwen3.8 27B	Spaced letters	0.34	0.71	0.16	0.09	0.54	0.05	0.10	0.39	0.07
	Digits	0.25	0.73	0.10	0.05	0.66	0.02	0.06	0.28	0.05
	Dependency tasks	0.55	0.79	0.28	0.47	0.71	0.27	0.46	0.58	0.37
Gemma 4 31B	Spaced letters	0.25	0.77	0.09	0.07	0.45	0.04	0.06	0.44	0.04
	Digits	0.21	0.76	0.07	0.01	0.52	0.01	0.02	0.24	0.01
	Dependency tasks	0.40	0.63	0.26	0.20	0.44	0.15	0.31	0.40	0.27
Aligned scoring										
Qwen2.5 14B	Spaced letters	0.35	0.51	0.27	0.21	0.41	0.16	0.19	0.38	0.15
	Digits	0.28	0.44	0.22	0.22	0.41	0.17	0.15	0.35	0.12
	Dependency tasks	0.35	0.54	0.26	0.33	0.47	0.26	0.35	0.49	0.28
Qwen2.5 32B	Spaced letters	0.25	0.46	0.18	0.11	0.34	0.08	0.07	0.35	0.05
	Digits	0.21	0.41	0.16	0.10	0.35	0.08	0.06	0.32	0.05
	Dependency tasks	0.34	0.52	0.25	0.31	0.51	0.23	0.32	0.45	0.26
Qwen3.8 27B	Spaced letters	0.19	0.44	0.14	0.06	0.36	0.04	0.08	0.30	0.07
	Digits	0.17	0.45	0.11	0.03	0.32	0.02	0.06	0.26	0.05
	Dependency tasks	0.40	0.57	0.29	0.26	0.51	0.18	0.32	0.45	0.26
Gemma 4 31B	Spaced letters	0.11	0.42	0.07	0.05	0.36	0.04	0.06	0.40	0.04
	Digits	0.08	0.34	0.06	0.00	0.10	0.00	0.01	0.21	0.01
	Dependency tasks	0.21	0.36	0.17	0.08	0.12	0.08	0.21	0.20	0.21

Table 16: The blind spot at $k = 2$, 3, and 4 separately (dependency tasks, four stronger models, $N \leq 64$): change at the post-marker and within-segment positions against the no-marker run at the same positions. $k = 3$ is otherwise excluded from the pooled analyses.

		Post-marker change			Within-segment change		
Task	Model	$k = 2$	$k = 3$	$k = 4$	$k = 2$	$k = 3$	$k = 4$
`copy_previous`	Qwen2.5 14B	−0.19	−0.18	−0.25	−0.14	−0.13	−0.10
	Qwen2.5 32B	−0.16	−0.12	−0.18	−0.13	−0.05	−0.01
	Qwen3.8 27B	−0.47	−0.41	−0.43	−0.33	−0.25	−0.21
	Gemma 4 31B	−0.33	−0.26	−0.26	−0.10	−0.10	−0.09
`running_xor`	Qwen2.5 14B	−0.17	−0.20	−0.13	−0.02	+0.08	+0.09
	Qwen2.5 32B	−0.22	−0.20	−0.19	+0.09	+0.12	+0.14
	Qwen3.8 27B	−0.07	−0.14	−0.10	+0.06	+0.11	+0.18
	Gemma 4 31B	−0.27	−0.28	−0.28	+0.11	+0.10	+0.10
`running_sum_local`	Qwen2.5 14B	−0.10	−0.17	−0.10	+0.01	+0.12	+0.14
	Qwen2.5 32B	−0.20	−0.20	−0.17	+0.12	+0.09	+0.12
	Qwen3.8 27B	−0.09	−0.17	−0.09	+0.12	+0.15	+0.15
	Gemma 4 31B	−0.43	−0.47	−0.45	+0.06	+0.08	+0.12
`sliding_window_max`	Qwen2.5 14B	−0.23	−0.18	−0.16	−0.11	−0.11	+0.00
	Qwen2.5 32B	−0.02	−0.04	−0.11	+0.08	+0.11	+0.10
	Qwen3.8 27B	−0.22	−0.25	−0.20	−0.01	+0.10	+0.13
	Gemma 4 31B	+0.01	+0.12	−0.14	+0.11	+0.04	+0.04

F Reset variants, pooled

Table 17 gives the pooled effect of markers on each dependency task in its original and reset forms; the per-position comparison is in Section 6.1.

Table 17: Reset variants, pooled over positions: Character accuracy Base → Bounds, Δ, and $|z|$, pooled as in Figure 5 (no-marker mean length-matched). Base differs between the original and reset columns because the reset prompt alone shifts no-marker accuracy; Section 6.1 therefore compares marker runs position by position. Originals use 20 to 80 trials per cell (Table 9); reset variants 20, except 80 at Qwen2.5-7B.

		Original (dependency intact)		Reset (dependency severed)	
Task	Model	Base→Bounds	Δ ($\lvert z\rvert$)	Base→Bounds	Δ ($\lvert z\rvert$)
`running_xor`	7B	0.401→0.403	+0.002 (0.1)	0.450→0.482	+0.032 (3.4)
	32B	0.746→0.798	+0.053 (6.2)	0.715→0.878	+0.163 (11.9)
	Qwen3.8 27B	0.646→0.742	+0.096 (9.5)	0.645→0.855	+0.210 (14.4)
	Gemma 4 31B	0.816→0.850	+0.034 (4.3)	0.816→0.928	+0.112 (10.2)
`running_sum_local`	7B	0.286→0.280	−0.006 (0.8)	0.277→0.348	+0.071 (12.4)
	32B	0.547→0.603	+0.056 (9.0)	0.508→0.660	+0.152 (12.1)
	Qwen3.8 27B	0.441→0.573	+0.131 (13.0)	0.419→0.683	+0.264 (19.4)
	Gemma 4 31B	0.754→0.775	+0.021 (3.2)	0.723→0.840	+0.117 (10.8)
`sliding_window_max`	32B	0.668→0.733	+0.065 (7.9)	0.677→0.816	+0.139 (10.9)
	Qwen3.8 27B	0.666→0.653	−0.013 (1.0)	0.640→0.824	+0.183 (16.4)
	Gemma 4 31B	0.708→0.783	+0.075 (6.3)	0.718→0.817	+0.099 (7.9)
`copy_previous`	7B	0.508→0.418	−0.090 (7.0)	0.565→0.553	−0.013 (1.4)
	32B	0.836→0.818	−0.018 (1.9)	0.874→0.792	−0.083 (4.6)
	Qwen3.8 27B	0.971→0.709	−0.262 (23.9)	0.942→0.836	−0.105 (6.4)
	Gemma 4 31B	0.917→0.832	−0.086 (6.5)	0.837→0.866	+0.029 (1.3)

G External tasks

Tables 18 and 19 and Figures 10 and 11 give the results on tasks we did not design (Section 6) and on Gemini.

Table 18: External position-local tasks, pooled across length and density. Base: no-marker accuracy averaged over lengths. Best-k: accuracy at the single best spacing (optimistic). Δ: marker runs minus the length-matched no-marker mean (Section 3). ProcBench-Substitute was regenerated after we made the vendored ProcBench generator's substitution table deterministic (it was drawn in a process-dependent order).

	Qwen2.5-7B			Qwen2.5-32B		
Task	Base	Best-k	Δ ($\lvert z\rvert$)	Base	Best-k	Δ ($\lvert z\rvert$)
Vowel/consonant tagging	0.438	0.435	−0.083 (2.4)	0.615	0.657	+0.012 (0.5)
ProcBench-Substitute	0.189	0.432	+0.096 (6.7)	0.390	0.757	+0.203 (10.8)
String copying	0.973	1.000	+0.009 (1.0)	0.993	1.000	+0.007 (2.0)

Table 19: Closed-model check and CLRS-Text. Gemini 3.5 Flash (3 trials per cell, letter and dependency tasks, pooled over all tested lengths, including N up to 256 for `cyclic_shift`) and Gemini 3.6 Flash (5 trials per cell, all 13 task–format variants, $N \leq 64$), with the no-marker mean length-matched. CLRS-Text tasks (Gemini 3.5) use token-level markers and scoring.

Task	Model	Base acc.	Bounds acc.	Δ ($\lvert z\rvert$)
`cyclic_shift`	Gemini 3.5	0.596	0.852	+0.256 (14.4)
	Gemini 3.6	0.737	0.925	+0.188 (9.2)
`random_permutation`	Gemini 3.5	0.561	0.846	+0.284 (8.4)
	Gemini 3.6	0.549	0.841	+0.291 (10.3)
`conditional_shift`	Gemini 3.5	0.453	0.628	+0.175 (3.1)
	Gemini 3.6	0.637	0.831	+0.195 (9.0)
`running_xor`	Gemini 3.5	0.882	0.913	+0.032 (0.4)
	Gemini 3.6	0.976	0.911	−0.065 (4.5)
`running_sum_local`	Gemini 3.5	0.918	0.803	−0.116 (5.2)
	Gemini 3.6	0.878	0.826	−0.051 (2.9)
`copy_previous`	Gemini 3.5	0.984	0.924	−0.060 (2.8)
	Gemini 3.6	1.000	0.970	−0.030 (4.7)
`sliding_window_max`	Gemini 3.6	0.849	0.840	−0.009 (0.5)
`cyclic_shift_spaced`	Gemini 3.6	0.992	1.000	+0.008 (1.5)
`random_permutation_spaced`	Gemini 3.6	0.970	1.000	+0.030 (2.7)
`conditional_shift_spaced`	Gemini 3.6	0.936	0.973	+0.037 (2.7)
`cyclic_shift_digits`	Gemini 3.6	1.000	1.000	−0.0002 (1.0)
`random_permutation_digits`	Gemini 3.6	0.981	0.996	+0.015 (2.0)
`conditional_shift_digits`	Gemini 3.6	0.985	0.996	+0.011 (1.8)
CLRS minimum	Qwen2.5-7B	0.133	0.183	+0.050 (3.3)
	Qwen2.5-32B	0.592	0.392	−0.200 (4.5)
	Gemini 3.5	0.997	0.654	−0.343 (7.3)
CLRS Kadane	Qwen2.5-7B	0.060	0.068	+0.008 (1.0)
	Qwen2.5-32B	0.186	0.122	−0.064 (2.0)
	Gemini 3.5	0.384	0.354	−0.030 (0.3)

On Kadane's algorithm, base accuracy at both Qwen scales is statistically indistinguishable from chance under a shuffle test, so we do not interpret its effects. Minimum is well above chance at Qwen2.5-32B ($z = 4.1$ from shuffled chance).

Gemini's tokenization, from the API's token counts. We cannot inspect Gemini's tokenizer, but every response reports the prompt's token count, and across the Gemini 3.6 grid the instruction is fixed

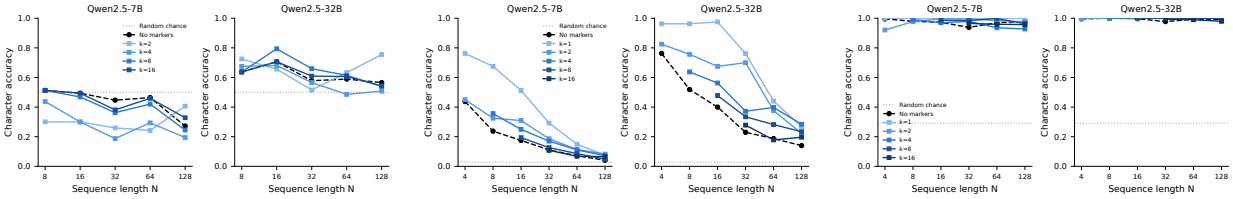


Figure 10: External tasks by marker density. Left: vowel/consonant tagging. Middle: ProcBench-Substitute. Right: string copying.

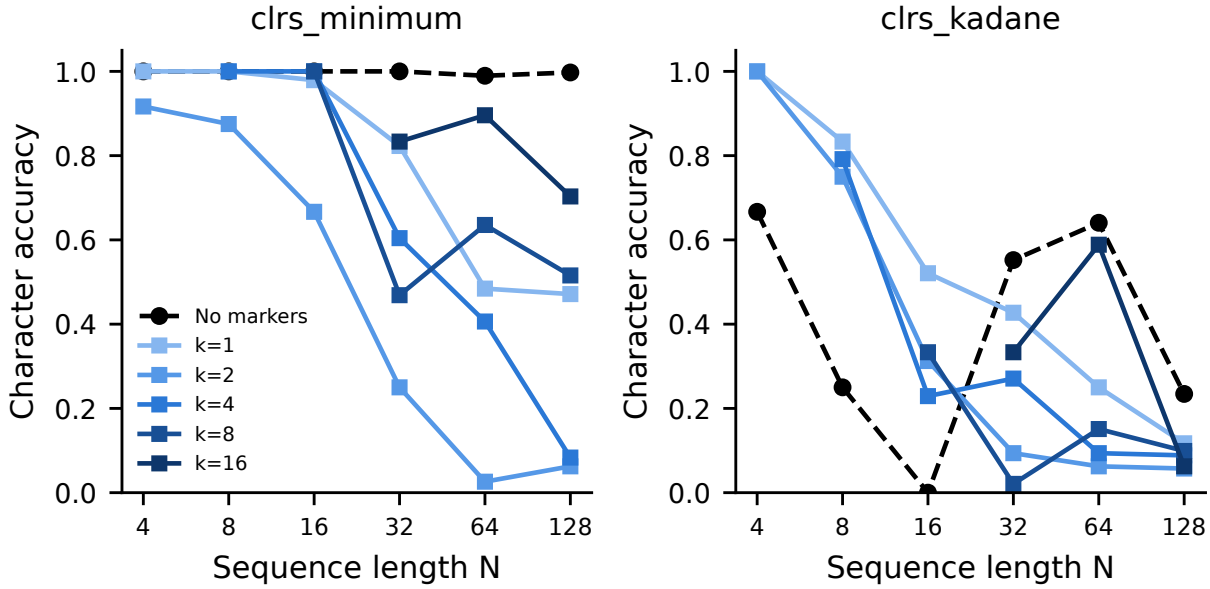


Figure 11: CLRS-Text tasks on Gemini 3.5 Flash by sequence length, one line per marker spacing k (3 trials per cell). Dashed: no markers.

while the input grows. Table 20 gives the prompt tokens added per added input character from $N = 32$ to $N = 64$, counting each marker as one token; the steps $4 \to 8$, $8 \to 16$, and $16 \to 32$ agree to within 0.04. In the spaced and digit formats and on the binary dependency tasks, every added character adds exactly one token and every marker exactly one more, at every length and spacing, so the one-token-per-character property holds at Gemini 3.6 as at the open models. In the letter format, letters without markers cost 0.46 tokens each (about 2.2 letters per token); a marker after every letter raises this to one token per letter, while markers every two, four, or eight letters leave it at 0.46 to 0.51, so at Gemini a sparse marker splits few of its letter tokens, which fits spacing beating markers there (Section 4). The counts fix how many tokens the characters occupy, not which characters share one; the open-model verification by inspection remains the primary one.

Table 20: Gemini 3.6 Flash: prompt tokens added per added input character, from the API's token counts, by input format and marker spacing k, at the step from $N = 32$ to $N = 64$, with each marker counted as one token (exact in the spaced, digit, and binary formats, where the count at spacing k exceeds the no-marker count by the number of markers at every N). A value of 1.00 is one token per character.

Input format	$k = 0$	$k = 1$	$k = 2$	$k = 4$	$k = 8$
Letters	0.46	1.00	0.50	0.51	0.46
Spaced letters	1.00	1.00	1.00	1.00	1.00
Digits	1.00	1.00	1.00	1.00	1.00
Binary digits (dependency tasks)	1.00	1.00	1.00	1.00	1.00

H Mechanism details

This appendix collects the attention analyses behind Section 7: the mechanism overview (Figure 12), example heads (Figure 13), the earlier head ablation (Figure 14), the head taxonomy (Table 22), and the 7B ablation (Table 21).

Table 21: Head ablation at Qwen2.5-7B, same design as Table 4. Marker gains are small at this scale, so shares are noisy; "–" marks cells with a gain below 0.03.

				Markers, with heads blocked			Share of gain removed	
Format	N	No markers	Markers	Top 10	Random 10	Top 10, sham	Top 10	Controls
Letters	32	0.17	0.24	0.23	0.24	0.24	14%	0%
Letters	64	0.14	0.16	0.14	0.15	0.16	–	–
Spaced letters	32	0.42	0.55	0.51	0.52	0.55	26%	$\leq$ 22%
Spaced letters	64	0.19	0.30	0.27	0.31	0.32	24%	0%
Digits	32	0.24	0.27	0.28	0.29	0.28	–	–
Digits	64	0.18	0.19	0.19	0.19	0.19	–	–

Table 22: The ten most marker-attending heads at Qwen2.5-32B, with percentiles within all heads on an induction test and a positional (previous-token) test.

Layer/Head	Marker attention	Induction pct.	Positional pct.
21/1	0.099	38.3	98.3
37/1	0.087	47.4	96.7
1/7	0.086	17.1	97.7
35/19	0.082	23.1	96.5
35/3	0.080	14.7	97.2
30/32	0.079	18.6	96.0
32/37	0.075	14.3	98.8
1/8	0.072	64.8	97.7
32/7	0.071	60.4	90.2
19/19	0.068	9.9	83.9
Mean		30.9	95.3

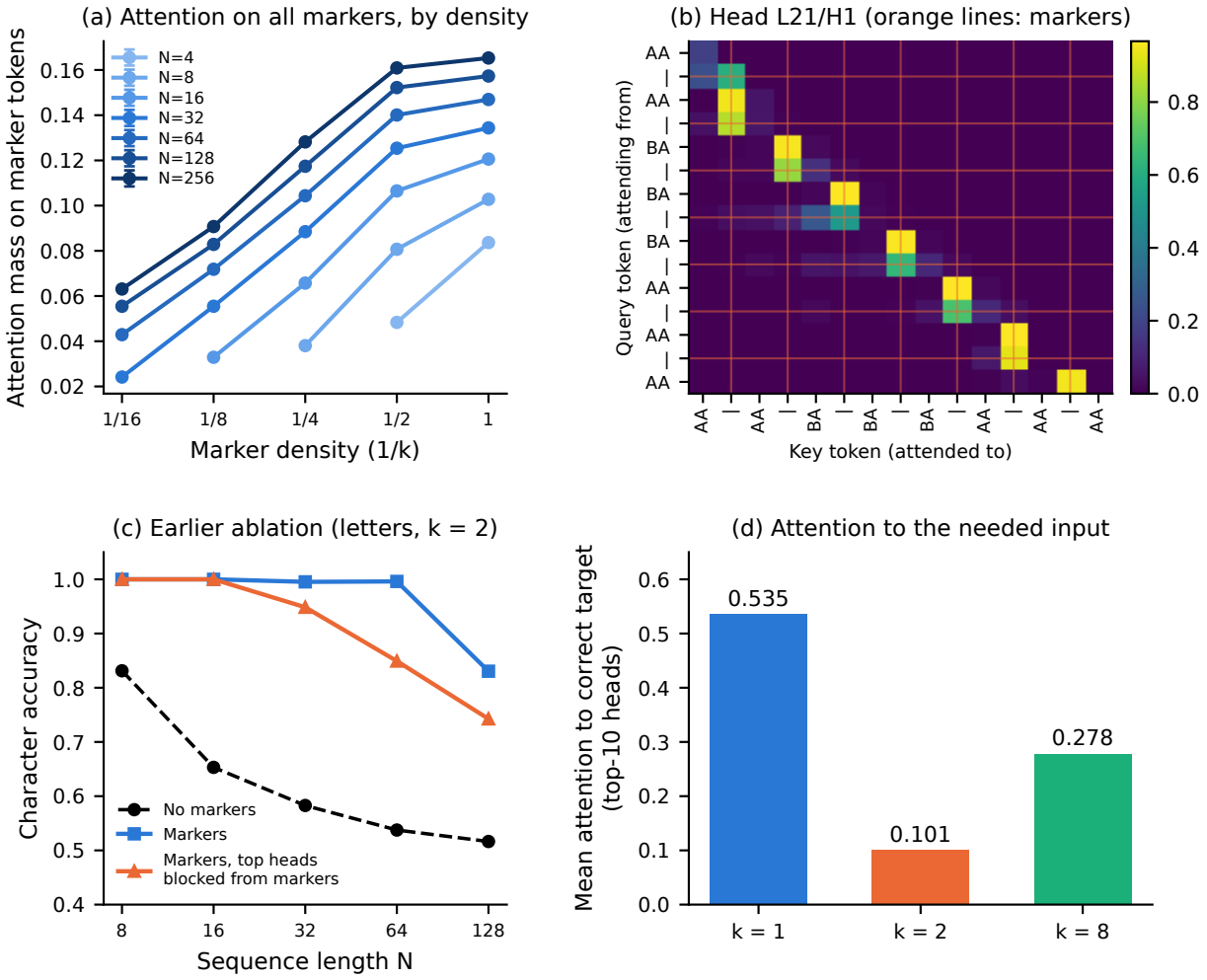


Figure 12: Mechanism (Qwen2.5-32B). (a) Total attention mass on all marker tokens rises with density at every length (`cyclic_shift`, $A = 2$); the number of markers rises faster, so this is not evidence that individual markers draw more attention than content tokens. (b) The most marker-attending head (layer 21, head 1) attends from each position to an adjacent marker. (c) Earlier ablation on contiguous letters ($k = 2$; Section 7): masking the ten most marker-attending heads' attention to markers removes a growing share of the benefit as N grows (0% at $N \leq 16$, 11% at 32, 32% at 64, 28% at 128). (d) `copy_previous` at $N = 32$: mean attention from each output position to the input token it needs (top-10 heads) is 0.535 at $k = 1$, 0.101 at $k = 2$, and 0.278 at $k = 8$, matching accuracy (Base 0.83; $k = 1$: 0.88; $k = 2$: 0.65; $k = 8$: 0.84).